

3.3.3 Separating mixtures

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 5			

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
A mixture contains two or more substances not chemically combined together. Mixtures can be separated by processes such as filtration, distillation, crystallisation and chromatography.	Students should be able to select methods from those given to separate simple mixtures. Suggested activity for TDA Compare the time needed to filter mixtures of water and calcium carbonate that has different particle sizes.	5.1.1.2	4.2.2.4
Outcome 6			
Paper chromatography can be used to separate mixtures and can give information to help identify substances. In paper chromatography a solvent moves through the paper carrying different compounds different distances.	Suggested activity for TDA Investigate the different colours in inks or food colours using paper chromatography.	5.8.1.3	4.2.2.4